Validating Data for a Linguistic Resource

Error analysis, agreement, and giving back

Francis **Bond**

Computational Linguistics Lab

Palacký University, Olomouc, Czechia

JPC3 — Improving the Czech Wordnet

November 2, 2026

Faculty of Arts

Overview

- Why be sure about the data
- Error analysis: beyond one number
- Why two annotators
- Cohen's κ, worked through
- Giving fixes back to ARASAAC, Wikidata, …

Roadmap

1. Why be sure?
2. Error analysis
3. Agreement
4. Cohen's kappa
5. Giving back

Why be sure?

- A wordnet is a **trusted source**
- Many tools read it; few check it
- An error you add, everyone inherits
- Once released, it is copied — hard to recall
- The Czech wordnet feeds OMW, sense tagging, AAC, …

The cost of one wrong link

- Project 4 links pictographs for **AAC** users
- A wrong synset tag $\rightarrow$ the wrong picture
- A child who cannot speak is handed the wrong word
- Here a lexical slip is not cosmetic — it misleads

Precision is not pedantry when someone relies on the data

Roadmap

1. Why be sure?
2. Error analysis
3. Agreement
4. Cohen's kappa
5. Giving back

Error Analysis

- Take a **random** sample (aim for 200)
- Judge each by hand: right or wrong?
- Precision = correct / judged
- But the single number is the least of it
- Group the mistakes into **types** — that is the payoff
- One real example per type, with a count

Running example: přechylování

Candidate: add a feminine form to a role-noun synset.

(1) Czech [ces]
učitel → *učitel-ka*
teacher.M teacher-F
“(male) teacher → (female) teacher”

- 1,919 role-noun synsets to work from (`noun.person`)
- Easy to generate; the question is which are **right**

Faculty of Arts

An Error Taxonomy

A sample of 200 generated feminine candidates:

Error type	Count	Example
Correct	171	*lékař* → *lékařka*
Coincidental suffix	12	*past* (not a base + *-ka*)
Wrong sense	9	feminine of a non-personal reading
Already present	8	lemma was in the synset

Precision 0.86 — but the four rows tell the next team where to look

Faculty of Arts

Roadmap

1. Why be sure?
2. Error analysis
3. Agreement
4. Cohen's kappa
5. Giving back

Why Two Annotators?

- One judge can be **consistently** wrong
- A second judge tests whether the scheme is learnable
- Agreement $\approx$ reliability of the guidelines
- You work in pairs — so measure it
- Report how you resolved the disagreements

Raw Agreement Is Not Enough

- “We agreed 85% of the time” sounds good
- But some agreement happens by **chance**
- Two coin-flippers agree half the time
- If almost everything is “correct”, agreeing is easy
- Correct for chance $\rightarrow$ Cohen’s κ

Roadmap

1. Why be sure?
2. Error analysis
3. Agreement
4. Cohen's kappa
5. Giving back

Cohen's Kappa

Agreement above what chance would give:

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

- p_o: observed agreement (the proportion you both called the same)
- p_e: expected agreement **by chance**, from each annotator's rates
- $\kappa = 1$ perfect; $\kappa = 0$ no better than chance

Faculty of Arts

A Worked Example

200 candidates, two annotators, good vs. bad:

	B: good	B: bad	total
A: good	130	20	150
A: bad	10	40	50
total	140	60	200

- $p_o = (130 + 40)/200 = 0.85$
- $p_e = 0.75 \times 0.70 + 0.25 \times 0.30 = 0.60$

$$\kappa = \frac{0.85 - 0.60}{1 - 0.60} = \mathbf{0.63}$$

Reading Kappa

- Rough scale (Landis & Koch, 1977): > 0.8 very good, 0.6–0.8 substantial
- Our 0.63 is **substantial** — the guidelines mostly work
- Low κ? Fix the guidelines, then re-annotate
- Beware: few categories or a skewed split inflate it
- Report κ *and* the raw table, never κ alone

Faculty of Arts

Roadmap

1. Why be sure?
2. Error analysis
3. Agreement
4. Cohen's kappa
5. Giving back

Faculty of Arts

Feedback Upstream

Every error you find is a **gift** to the source.

- ARASAAC: report wrong synset tags; send Czech keywords back
- Wikidata: add or fix the ILI (P5063) and WordNet id (P8814)
- Global Symbols / ARAWN: contribute the matches you validated
- CILI: report synsets with no ILI; DeriNet: report gaps
- PDT: the unmapped residue is data for the maintainers

The Virtuous Cycle

- Open data gets better only when fixes **flow back**
- A pull request or an issue beats a private fix
- Cite what you used; credit who made it
- Release under an open licence, loadable with `wn`
- Next year's class starts from your cleaner data

Every new wordnet makes the network richer

Faculty of Arts

Conclusion

- Extraction is scaffolded — **validation** is the work
- Sample, judge, and report precision
- Categorise the errors; the taxonomy guides the next step
- Two annotators, and κ with the table
- Push every fix back upstream

Thanks

- Validate carefully — others will trust your data
- Actually look at your data!
- Code and template: the JPC3 repository

Questions welcome — and so are pull requests

References I


Cohen, J. (1960). A coefficient of agreement for nominal scales. *Educational and Psychological Measurement* 20(1), 37–46.


Landis, J. R. & Koch, G. G. (1977). The measurement of observer agreement for categorical data. *Biometrics* 33(1), 159–174.


Artstein, R. & Poesio, M. (2008). Inter-coder agreement for computational linguistics. *Computational Linguistics* 34(4), 555–596.